

Assignment 3 – RNA-Seq Guided Replication Using Galaxy and GitHub

Group 1 – Heat Stress

Group Members and Roles

Member	Suggested Role	Notes
Rhealyn F. Alama	Galaxy lead / Documentation lead	Operated the designated Galaxy account (alama_rhealyn_)
Nesie D. Calipas	Literature lead	Assisted with paper selection and methods review
Mary Rose V. Ferrer	Data lead	Assisted with accession/data location
Jasmin Grace T. Bullanday	Interpretation lead	Assisted with comparison to published results

Part 3. Study Approval Information

Item	Group Information
Full title of the paper	Transcriptomic Profiling of <i>Paulownia elongata</i> in Response to Heat Stress
Authors	Neerja Katiyar, Niveditha Ramadoss, Dinesh Gupta, Suman B. Pakala, Kerry Cooper, and Chhandak Basu
Year	2021
Journal	Plant Gene, Vol. 28, 100330
Organism	<i>Paulownia elongata</i>
Tissue or cell type	Leaves
Assigned stress topic	Heat stress
Control condition	25 °C, 12h light/12h dark photoperiod
Treatment/stress condition	40 °C for 24 hours, same photoperiod
Number of biological replicates	3 per condition (3 control, 3 heat-stressed)
RNA-seq repository	NCBI SRA / GEO
BioProject / Study accession	PRJNA485845 as printed in the paper (this accession returns no items on NCBI); the correct linked BioProject on SRA is PRJNA488054. SRA study: SRP158908
Run accession numbers the group plans to use	Control: SRR7758331 (rep1), SRR7758332 (rep2), SRR7758333 (rep3) Heat-treated: SRR7758328 (rep1), SRR7758329 (rep2), SRR7758330 (rep3)
Reference genome accession	No public reference genome exists for <i>P. elongata</i> . The group substituted the congeneric species genome <i>Paulownia fortunei</i> (Cao et al., 2021, Molecular Plant), assembly GCA_019321725.1 / JAGSPU000000000

Item	Group Information
Annotation source	GFF3 annotation packaged with the <i>P. fortunei</i> NCBI genome assembly (downloaded together with the genome FASTA via the Galaxy NCBI Datasets Genomes tool)
Approximate size of the selected RNA-seq files	~175 Mb per run (download size), ~3–5 million paired-end spots per sample
Short description of the authors' RNA-seq analysis pipeline	De novo transcriptome assembly with Trinity (v2.0.6) (no reference genome existed for <i>P. elongata</i>) → contamination screening with DIAMOND/BLAST and MEGAN6 → completeness assessment with BUSCO → functional annotation with Blast2GO/OmicsBox → transcript quantification with Salmon → differential expression with edgeR (FDR < 0.05, log2FC > 1) → GO/KEGG enrichment with OmicsBox

Part 4. Methods Identified From the Paper

- **Sequencing platform:** Illumina MiSeq (MiSeq Reagent Kit v3)
- **Single-end or paired-end:** paired-end, 76 bp reads
- **Read quality-control method:** not explicitly named beyond standard QC prior to assembly
- **Adapter or quality trimming:** not explicitly detailed in the Methods section
- **Reference genome/transcriptome:** none available; authors built a de novo transcriptome with Trinity (v2.0.6, default settings)
- **Read alignment/pseudo-alignment:** Salmon (pseudo-alignment/quantification against the de novo transcriptome)
- **Gene counting/transcript quantification:** Salmon read quantification, analyzed with the Bioconductor package edgeR
- **Differential expression software:** edgeR, using an Exact Test
- **Important parameters/thresholds:** FDR < 0.05 and |log2 fold change| > 1
- **Criteria for differentially expressed genes:** FDR < 0.05 and |log2FC| > 1 (2,797 genes upregulated; 1,638 genes downregulated; 4,435 total DEGs reported by the authors)

Part 5–6. Galaxy Analysis History and Data Import

Galaxy history name: Group1_HeatStress_RNASeq_Assignment3

Dataset renamed as	Run accession	Condition	Replicate
Control_R1_rep1 / Control_R2_rep1	SRR7758331	Control	1
Control_R1_rep2 / Control_R2_rep2	SRR7758332	Control	2
Control_R1_rep3 / Control_R2_rep3	SRR7758333	Control	3
Heat_R1_rep1 / Heat_R2_rep1	SRR7758328	Heat	1
Heat_R1_rep2 / Heat_R2_rep2	SRR7758329	Heat	2
Heat_R1_rep3 / Heat_R2_rep3	SRR7758330	Heat	3

Part 7. RNA-Seq Quality Control

FastQC was run on all 12 FASTQ files (R1 and R2 for each of the 6 samples), and MultiQC was used to summarize the reports.

QC metric	Observation across all 12 files
Number of reads	Ranged from roughly 1.3 to 5 million read pairs per sample (varies by run)
General sequence quality	High throughout — per-base Phred scores mostly in the Q32–Q38 range
Adapter contamination	None detected in any sample
Over-represented sequences	None flagged as a concern
Notable quality warning	"Per Base Sequence Content" failed on all 12 files — a common RNA-seq artifact caused by non-random priming during library prep (fragment-start base-composition bias), not a sign of contamination or low-quality data

Part 8. Trimming Decision

Because no adapter contamination was detected and per-base quality was high across all samples, the group decided not to trim the reads. The "Per Base Sequence Content" failure was interpreted as an expected RNA-seq priming-bias artifact rather than a quality problem requiring trimming, and reads proceeded directly to mapping.

Part 9. Reference Genome and Annotation

Item	Detail
Genome used	<i>Paulownia fortunei</i> (GCA_019321725.1 / JAGSPU000000000), obtained via the Galaxy NCBI Datasets Genomes tool
Reason for substitution	No reference genome exists for <i>P. elongata</i> (the original authors also lacked one and used a de novo Trinity assembly instead). <i>P. fortunei</i> is the closest congeneric species with a published, annotated, chromosome-level genome (Cao et al., 2021, Molecular Plant).
Annotation file	GFF3 annotation downloaded together with the genome FASTA (same NCBI Datasets Genomes job)
Assembly/annotation version	GCA_019321725.1 (<i>P. fortunei</i> nuclear genome, 511.6 Mb, 20 pseudo-chromosomes, 31,985 protein-coding genes)

Part 10. Read Mapping

Each of the 6 paired-end samples was mapped to the *P. fortunei* genome FASTA using HISAT2 (default parameters, paired-end mode) in Galaxy.

Sample	Total reads	% mapped	% uniquely mapped	Notes
Control_rep1 (SRR7758331)	4,956,271	88.92%	74.71%	—
Control_rep2 (SRR7758332)	3,950,092	88.98%	71.26%	Highest total reads; lowest unique-mapping rate
Control_rep3 (SRR7758333)	3,147,060	89.24%	76.07%	—
Heat_rep1 (SRR7758328)	1,349,099	90.92%	78.31%	Read depth much lower than other samples
Heat_rep2 (SRR7758329)	3,215,196	88.51%	72.70%	Lowest overall alignment rate
Heat_rep3 (SRR7758330)	3,441,734	92.74%	80.25%	Highest unique mapping rate

Part 11. Gene Counting

featureCounts was run on each of the 6 HISAT2 BAM outputs against the *P. fortunei* GFF3 annotation, producing a count table with one column per sample.

Item	Detail
Counting tool	featureCounts (Galaxy)
Annotation used	<i>P. fortunei</i> GFF3 (same file used for mapping reference)
Samples counted	All 6 (3 control, 3 heat-stressed)
Output	Gene-level count table, one column per sample, used directly as DESeq2 input

Part 12. Differential Gene Expression (DESeq2)

DESeq2 was run in Galaxy on the featureCounts gene count table, with the 3 control replicates set as the reference condition against the 3 heat-stressed replicates.

Output	What it showed
PCA plot	PC1 explained 55% of variance and PC2 explained 27%; samples separated into two clear groups by condition (Control vs. Heat)
Sample-to-sample distance heatmap	Generated alongside the PCA to confirm replicate clustering by condition
Dispersion estimates plot	Gene-wise, fitted, and final dispersion estimates plotted against mean normalized counts, following the expected DESeq2 shape
Histogram of p-values	Distribution of raw p-values for the Heat vs. Control comparison
MA-plot	Log fold change plotted against mean normalized counts for the Heat vs. Control comparison

Simple interpretation of DESeq2 output

Result	Simple meaning
Positive log ₂ fold change	Gene expression is higher in the heat-stress condition
Negative log ₂ fold change	Gene expression is lower in the heat-stress condition
Adjusted p-value < 0.05	Commonly treated as evidence the expression difference is statistically significant
Adjusted p-value ≥ 0.05	The analysis does not provide strong statistical evidence of differential expression

Item	Value
Number of genes tested	31,673
Number of significantly differentially expressed genes (padj < 0.05)	1,481
Genes with positive log ₂ fold change (up in heat)	827
Genes with negative log ₂ fold change (down in heat)	654
Adjusted p-value / FDR threshold used	0.05

Part 13. Genes Selected for Biological Interpretation

Gene name / ID	log2FC	Adjusted p-value	Up/down	Known or predicted function	Possible connection to heat stress
Tonoplast dicarboxylate transporter (TR36547 c0_g1_i1)	11.26	1.23E-08	Up	Vacuolar malate carrier; moves malic acid from vacuole to cytoplasm for catabolism	May prevent malate over-accumulation seen when temperature rises, helping regulate metabolite balance under heat
Heat shock protein 82 / HSP83-90 family (TR20200 c0_g2_i1)	10.74	9.52E-12	Up	Molecular chaperone; assists in refolding/stabilizing heat-denatured proteins	Classic heat-shock protein; protects native proteins from denaturation during thermal stress
Triacylglycerol (TAG) lipase (TR54032 c0_g2_i1)	10.30	5.47E-07	Up	Hydrolyzes long-chain triacylglycerols, releasing carbon skeletons/energy	Likely drives lipid remodeling to reduce toxic lipid intermediates that damage membranes under heat
Cystathionine beta-lyase (TR2916 c0_g1_i2)	10.01	3.13E-07	Up	Key enzyme in methionine biosynthesis pathway	Methionine feeds polyamine synthesis, which supports abiotic stress tolerance
50S ribosomal protein L1, chloroplastic (TR48060 c0_g1_i1)	-11.06	0.0006	Down	Structural component of the chloroplastic ribosome	Downregulating chloroplastic ribosomal genes is thought to redirect resources toward transcriptional activation of heat-responsive genes

Part 14. Comparison With the Original Study

- **Did your group recover the same general biological response described by the authors?**
 - The PCA already shows a clear separation between control and heat-stressed samples, consistent with the authors reporting thousands of differentially expressed genes under heat stress; a direct gene-level comparison requires the group's finalized DEG table.
- **Were any of the genes highlighted in the paper also identified in your analysis?**
 - Tonoplast dicarboxylate transporter and Heat shock protein .
- **Which results were similar?**
 - The overall scale and direction are similar in proportion, roughly 1.5-3x more upregulated than downregulated genes in both analyses, and a clean PCA/clustering separation by condition in both studies.
- **Which results were different?**

- The absolute number of DEG's is much smaller in the group's re-analysis (1,481 vs 4,435), which is expected given a different reference and a different statistical pipeline.

- **Did you use exactly the same software as the authors?**

- Because the RNA sequence for *P. elongata* is not publicly available, the group used HISAT2 for mapping, featureCounts for gene counting, and DESeq2 for differential expression, whereas the authors used a de novo Trinity assembly, Salmon quantification, and edgeR.

- **Did you use the same reference genome and annotation versions?**

- There is no *P. elongata* genome that exists as a publicly available reference. The group mapped to the congeneric *P. fortunei* genome (GCA_019321725.1) instead of building a de novo transcriptome.

- **Did you analyze all samples or only a subset?**

- All 6 samples (3 control, 3 heat-stressed) were analyzed. The full published sample set for this comparison.

- **Could differences in software versions, parameters, sample number, or reference files explain differences in the results?**

- Yes, mapping reads to a different though closely related species' genome instead of a de novo transcriptome assembled from the same species can shift which genes are detected, how reads are assigned, and normalization, and the aligner/counter/DE-testing software differs entirely from the original pipeline.

- **What did this exercise teach you about reproducibility in molecular biology?**

- This exercise showed that reproducibility depends on more than following the same general workflow; exact matches require the same raw data, reference files, software, and parameters, which are not always available. Using a related species' genome instead of a de novo assembly changed the number and identity of genes detected even though the overall heat-stress response pattern still matched the published study.

Part 16. GitHub Repository

Repository: https://github.com/rhealynalama-bio/RNASeq_Group-1_HeatStress

Part 17. Comparison Table: Original vs. Re-analysis

Analysis step	Original authors	Group's Galaxy re-analysis
RNA-seq dataset	6 samples (3 control, 3 heat), Illumina MiSeq, paired-end, 76 bp	Same 6 SRA runs (SRR7758328–333), same platform/read type
Quality control	Not explicitly detailed beyond standard pre-assembly QC	FastQC + MultiQC on all 12 files; high per-base quality, no adapters, expected priming-bias artifact
Trimming	Not explicitly detailed	None applied (not needed based on QC)
Reference genome	None available for <i>P. elongata</i> (de novo assembly used instead)	<i>P. fortunei</i> genome, GCA_019321725.1
Annotation	N/A (de novo transcriptome used as reference)	<i>P. fortunei</i> GFF3
RNA-seq aligner	Not applicable — pseudo-alignment via Salmon against the de novo transcriptome	HISAT2 (genome alignment)
Gene counting	Salmon quantification	featureCounts
Differential expression	edgeR (Exact Test)	DESeq2
Significance threshold	$FDR < 0.05$, $ \log_2FC > 1$	$padj < 0.05$
Main genes/pathways identified	HSP83/HSP90 and other heat shock proteins, tonoplast dicarboxylate transporter, TAG lipase upregulated; chloroplastic ribosomal proteins, cell-cycle/DNA replication genes downregulated	Heat shock protein 82 (HSP83/HSP90), tonoplast dicarboxylate transporter, TAG lipase, and cystathionine beta-lyase upregulated; chloroplastic ribosomal protein (50S L1) downregulated

Part 18. Molecular Biology Understanding Questions

- **1. Why can RNA-seq be used to study gene expression?**
 - RNA-seq sequences the transcripts a cell is actively producing, so the number of reads mapping to a gene reflects how much that gene is being transcribed under a given condition.
- **2. What is the difference between the genome FASTA and RNA-seq FASTQ files?**
 - The genome FASTA is the fixed DNA sequence (with only headers and nucleotide sequence); the RNA-seq FASTQ contains short sequenced reads from expressed transcripts plus a per-base quality score for every read.
- **3. Why is a genome annotation needed for gene-level expression analysis?**
 - The annotation (GTF/GFF3) marks where genes and exons are located on the genome, so mapped reads can be assigned to the correct gene for counting.
- **4. What does a high number of reads for a gene generally suggest?**
 - That the gene is being transcribed at a high level (highly expressed) under the sampled condition, relative to genes with fewer reads.
- **5. Why should biological replicates be included?**

- To distinguish a real treatment effect from normal biological variation between individual plants/organisms, and to allow statistical testing of differential expression.
- **6. What is an upregulated gene?**
- A gene whose expression is higher in the treatment/stress condition than in the control (positive log₂ fold change).
- **7. What is a downregulated gene?**
- A gene whose expression is lower in the treatment/stress condition than in the control (negative log₂ fold change).
- **8. Why is normalization needed when comparing RNA-seq samples?**
- Samples can have different total sequencing depth and RNA composition; normalization adjusts counts so that expression levels are comparable across samples rather than reflecting differences in how deeply each sample was sequenced.
- **9. Why is adjusted p-value commonly used instead of only the raw p-value?**
- Testing thousands of genes at once creates many chances for false positives by chance alone; the adjusted p-value (FDR) corrects for this multiple-testing problem.
- **10. Why might your results differ from those reported by the original authors?**
- Different reference (a related species' genome vs. a de novo transcriptome built from the same species), different aligner/quantification/DE-testing software, and possible differences in parameters can all shift which genes appear significant.
- **11. How can changes in gene expression help an organism respond to environmental stress or infection?**
- By turning on protective genes (e.g., heat shock proteins, chaperones) and turning down non-essential processes (e.g., ribosome biogenesis, DNA replication), the organism reallocates resources toward survival and damage control.
- **12. Does increased RNA abundance always mean that the amount of functional protein also increases?**
- No, mRNA levels do not always match protein levels, because translation efficiency, protein stability/degradation, and post-translational regulation also control how much functional protein is ultimately produced.